

Assignment 4: Style Transfer

Model Setup

For this assignment, a pretrained network was used to extract feature representations from an image in order to replicate the style of that image with another's content. Specific convolutional layers were selected to represent content and style for the output image – the conv1_1, conv2_1, conv3_1, conv4_1, and conv5_1 layers represented the style, while the conv4_2 layer represented the content. Due to the use of two input images, the total loss was determined by finding both the content loss and the style loss. Content loss was set to be the mean squared error between the feature representations of the target image and the content image at layer conv4_2, i.e. how similar the shapes and main features of the two images were to each other. Style loss was computed using Gram matrices, which capture correlations between feature maps, indicating style features such as line width, brushstrokes, and color choice. The Gram matrix was calculated by reshaping the feature tensor and multiplying it by its transpose. Style loss was calculated across multiple layers and weighted to emphasize different levels of texture representation, focusing more on earlier layers. The total loss equation was defined as:

Total Loss = $\alpha \cdot \text{Content Loss} + \beta \cdot \text{Style Loss}$ where α = content weight, and β = style weight. It should be noted that style loss was generally a much larger part of the equation than content loss, due to the larger amount of layers, rendering it more influential in the final product. Finally, the target image was initialized as a copy of the content image and optimized using the Adam optimizer.



Figure 1: The content image



Figure 2: The style image

Experimentation

All experiments used the same content and style images to ensure consistency, as can be seen in Figures 1 and 2. The following hyperparameters were varied: style weight, content weight, learning rate, and number of optimization steps. The style weight was initially set to 136, the content weight to 1, the learning rate to 0.003, and the step number to 2000. Each experiment was evaluated using both total loss and visual inspection of the resulting image. The first parameter tested was style weight - initially set at a value of $1e6$ - resulting in Figure 4.

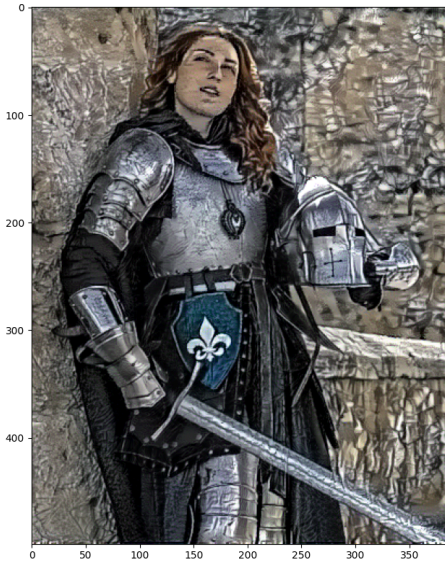


Figure 3: Style weight of 100

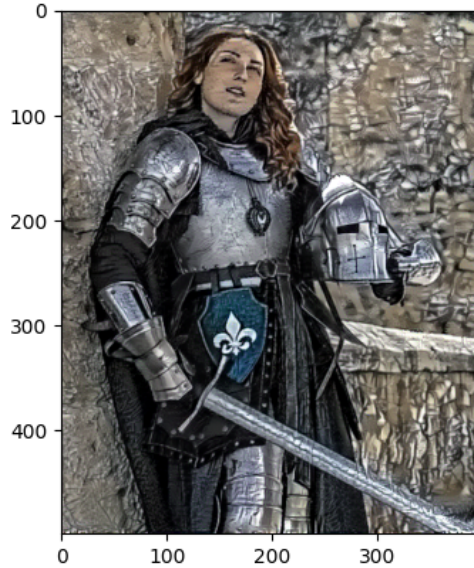


Figure 4: Style weight of $1e6$

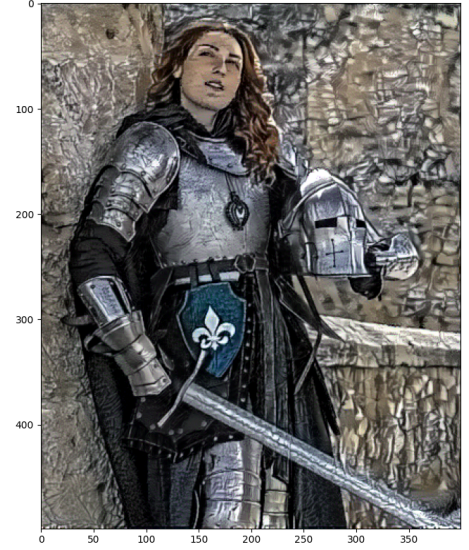


Figure 5: Style weight of $1e9$

The style weight was also tested at a value of 100 and $1e9$, resulting in Figures 3 and 5, respectively. As can be seen, there is not a great deal of difference in all three images – they all show a much more limited color palette, as well as clear changes to the background. The color change makes sense, as the style the model is trying to mimic is black and white, so the warmer colors of red, tan, and brown become subdued. As for the background, it appears to mimic the same background style as the style image, but with worse results as that image's background largely consists of clouds far in the distance. It would be interesting to test two images that are composed exactly the same to see if the background would be improved. The clearest indicator of the style transfer, in my opinion, is the helmet in these three images. The helmet in Figure 3 depicts it in a style much more similar to the style image, and in Figure 5 the helmet itself looks almost as if it was drawn with pen strokes. The loss values on the other hand were much different. The initial style weight of $1e6$ resulted in a total loss of around $4e6$, changing the weight to $1e2$ resulted in a much lower total loss of 423, and finally increasing the style weight to $1e9$ resulted in a total loss of $4e9$. This would seem to indicate that the smaller style weight results in a much better or more accurate output than a larger style weight, however the images seem very similar. At the end of the day, this shows an inherent problem with judging the quality of a model such as this – we cannot simply rely on the loss value to do so, and must also rely

on our opinion of the images. The differences between these images are fairly subtle, and it is up to personal interpretation if the style is mimicked accurately. Following this, the second parameter tested was content weight.

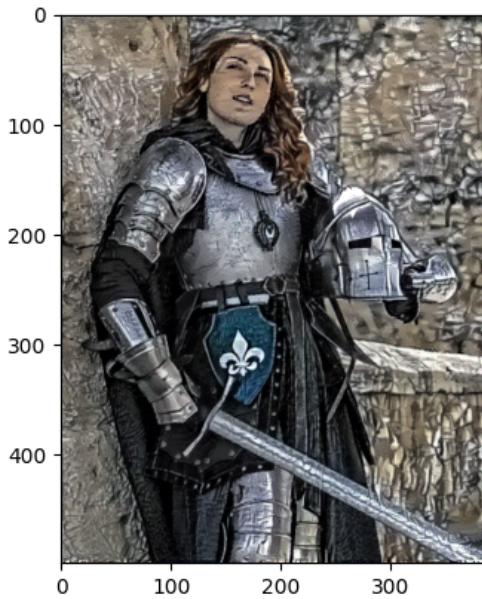


Figure 6: Content weight of 0.001

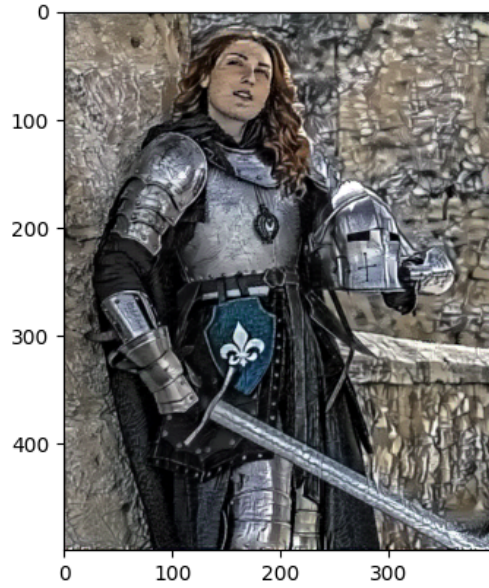


Figure 4 (again): Content weight of 1



Figure 7: Content weight of 100

As can be seen in Figures 6, 4, and 7, the results of modifying the content weight are similar to changing the style weight – all three images are very similar despite the magnitude of change. While one might argue the colors are brighter and more varied for Figure 7, it is a close thing. The loss values are also different than one might expect, as the initial content weight of 1 (Figure 4) resulted in - as mentioned above - a loss of $4e6$, changing the content weight to 0.001 resulted in the same loss value of $4e6$ as did increasing the content weight to 100. This was also mentioned above: the style weight has a much larger impact on the total loss than the content weight. This led to a realization that since the weights are opposing each other in the total loss equation, increasing just the content weight to 100 operated much the same as reducing the style weight to $1e4$. Due to this, both values should be changed more in relation to each other to see differences.

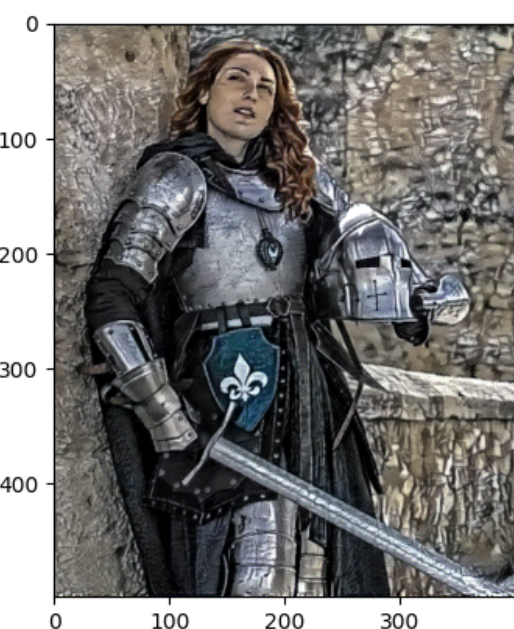


Figure 8: Equal content and style weight

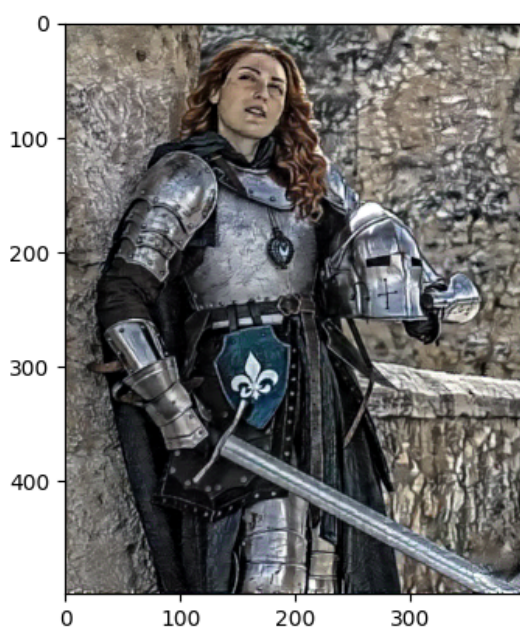


Figure 9: Bigger content than style

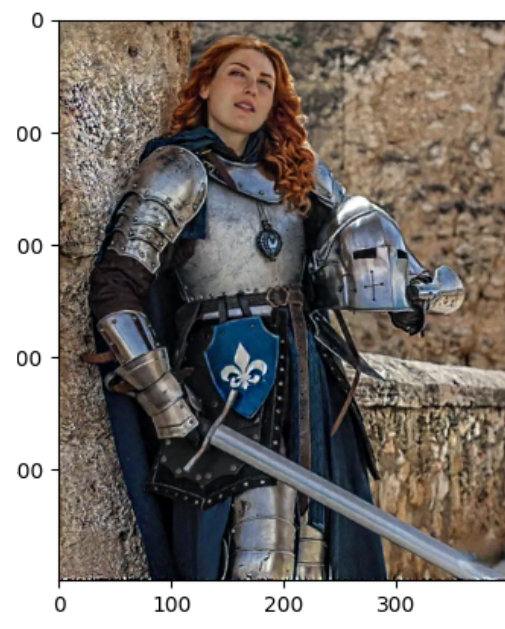


Figure 10: Much bigger content than style

After testing in this way, the differences were much easier to see. As Figure 8 indicates, setting the content weight and style weight to the same value (100) still did not change the image greatly, as the content loss was still larger due to the multiple layers. These settings also resulted in a total loss of only 770, the second-best value obtained so far. This is even evident in Figure 9, where the content weight was set to $1e3$ and the style weight was set to 100, however the loss was much larger: $4e9$. The big difference can only be seen in Figure 10, where the content weight was set to $1e9$, the style weight was set to 100, and the total loss found was $2e4$. As can be seen, the warmer browns, tans, and reds are completely visible, and the blue of the sigil is much deeper. We can also still see evidence of the style in the way the background has started to replace complex texture with linear designs, and the helmet still has sharp lines reminiscent of the style image. It's interesting that Figure 9 had the largest loss value by a significant factor of the three in this batch, but I believe it makes sense that increasing either the content or loss ratio to one extreme would allow the model to specialize towards that extreme and limit the loss. Learning rate was the next hyperparameter to be tested.

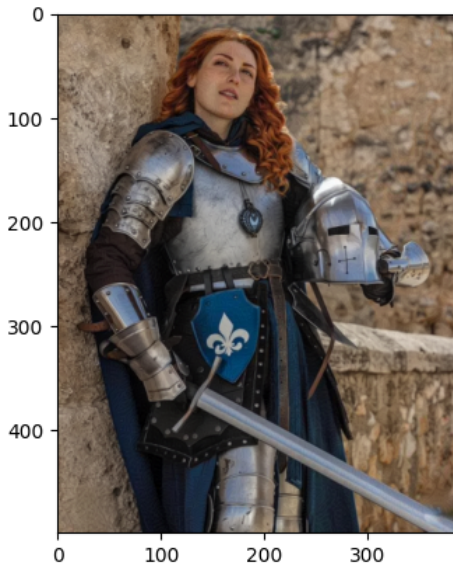


Figure 11: Learning rate of $3e-5$

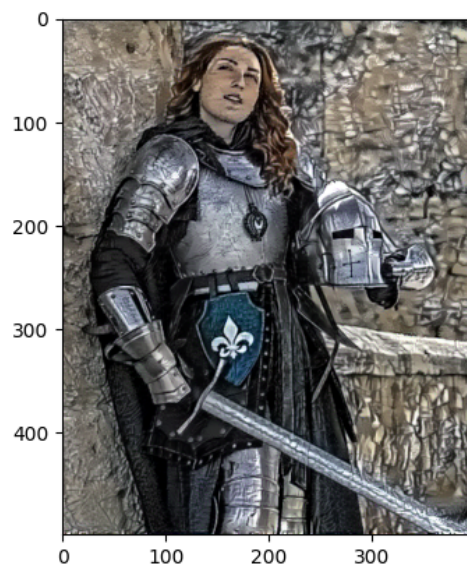


Figure 4 (again): Learning rate of 0.003

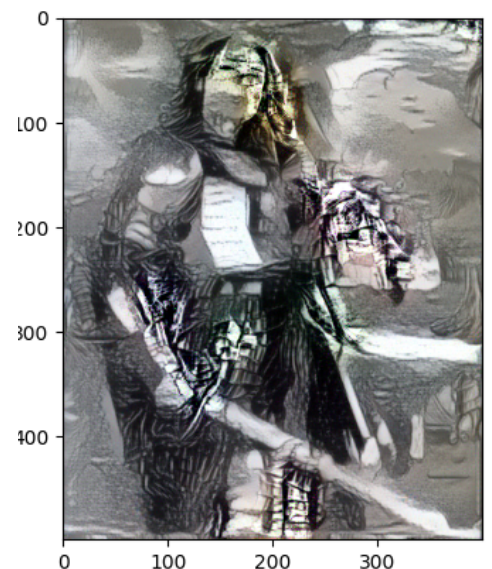


Figure 12: Learning rate of 0.3

Modifying the learning rate hyperparameter resulted in the three most distinct images in a batch so far. The differences here are clear; the learning rate of $3e-5$ did not have enough time to learn the style and so has Figure 11 is an image very similar to the input, the standard learning rate we've observed in Figure 4's attempt results in an interesting middle ground, and the very fast learning rate of 0.3 - as shown in Figure 12 - changes the image drastically. Figure 12 very clearly borrows the style at the expense of the content; the face is missing, the background appears to be entirely different and farther away, and almost all color is gone. What's interesting is that all three of these images have relatively high loss values. The very slow learning rate of $3e-5$ had a total loss of $5e8$ - which can make sense if we consider the style loss must be very high - while the very fast learning rate of 0.3 had a total loss of $5e6$. These, compared to the aforementioned $4e6$ loss of the standard Figure 4 image, seem to indicate that the learning rate is another balance between the content and style. It appears that the faster the learning rate, the more the model favors the style rather than the content, possibly due to the higher weights and larger amount of layers. Following this, the number of steps was the hyperparameter to focus on.

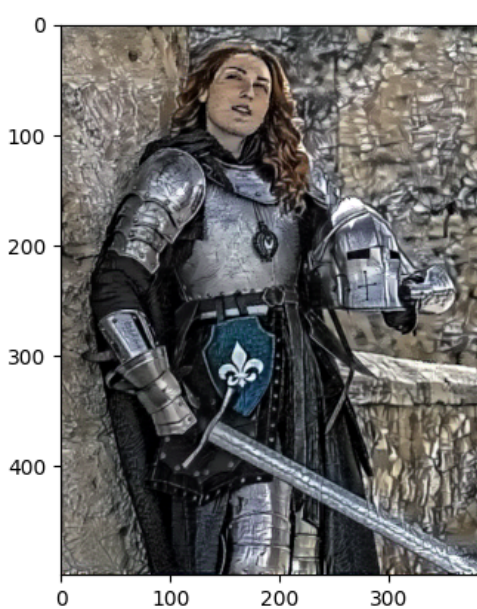


Figure 4 (again): 2000 steps

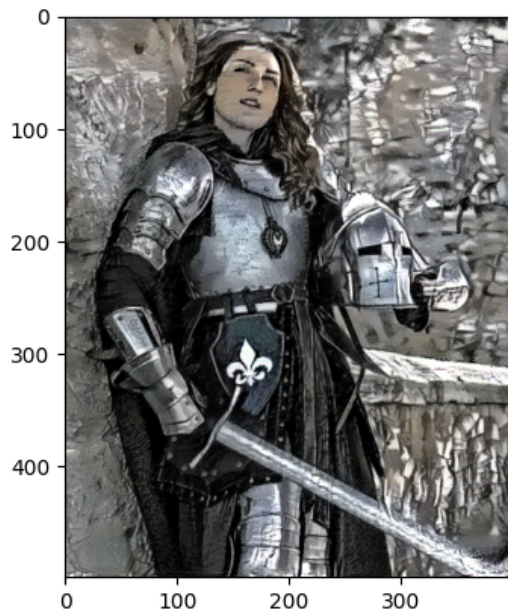


Figure 13: 5000 steps

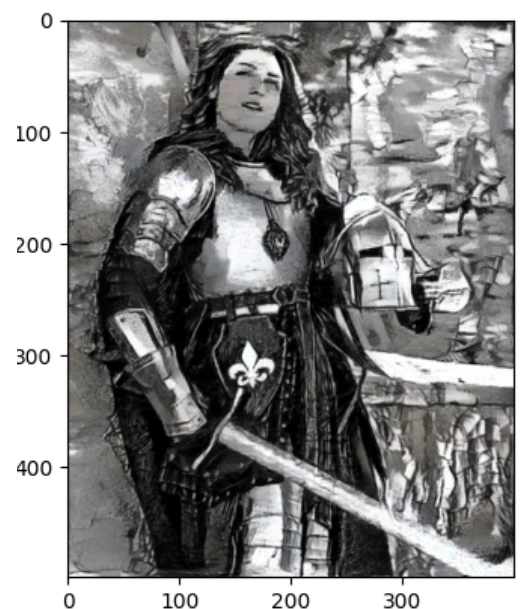


Figure 14: 20000 steps

While the learning rate had the most drastic differences between the three images in the batch, the step count hyperparameter appeared to produce the best qualitative batch to this point. As can be seen in Figure 14, 20000 steps were able to blend the style and content very effectively, resulting in a black-and-white image that not only was largely 2-D, but also replicated the content background in the correct style. This is also reflected in the total loss values; 20000 steps resulted in a value of $1.4e5$, while the 2000 and 5000 step methods both had a loss value of $4e6$. It's uncertain at what point the amount of steps results in diminishing returns, as no testing was done beyond 20000 steps due to the apparent runtime, but I would think 20000 is reaching a point close to it. Finally, once all the hyperparameters were tested, an attempt was made to modify each hyperparameter to a specific end.

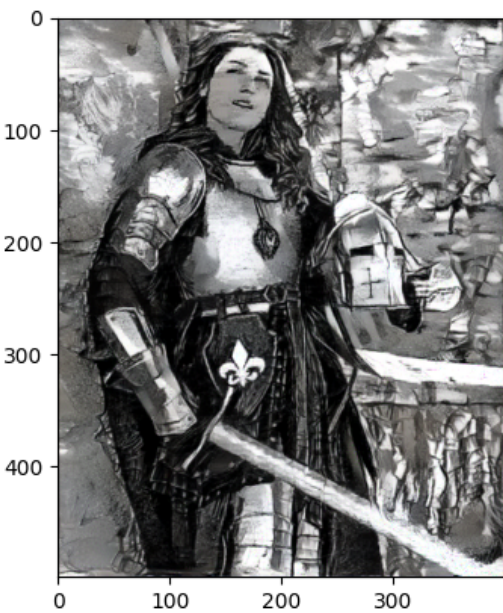


Figure 15: Smallest loss

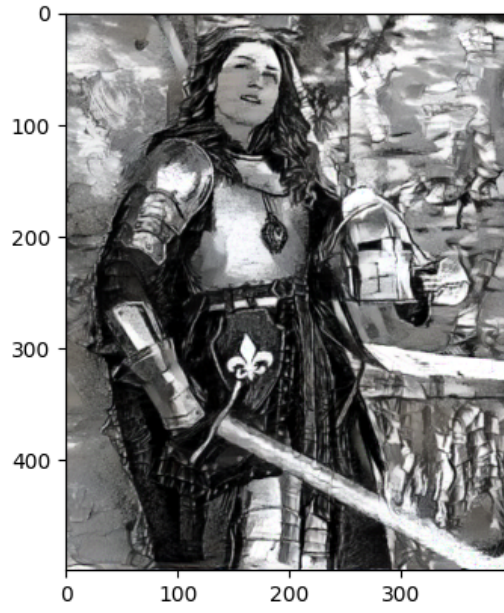


Figure 16: High steps, high style weight

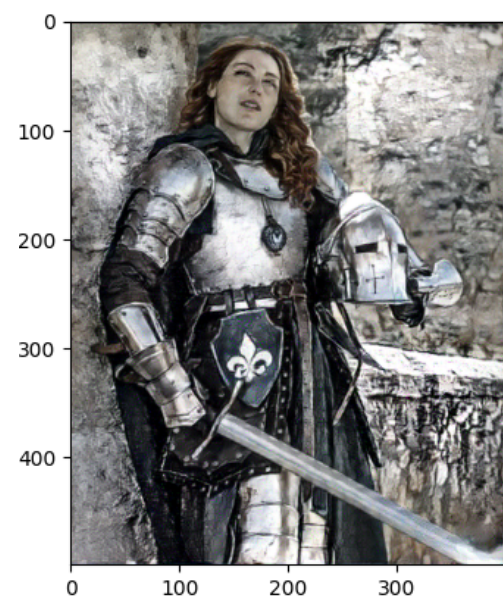


Figure 17: High content to style ratio

The first goal was to find the minimum amount of loss that could be achieved. This was done by examining each configuration of hyperparameters to see which ones limited the total loss the most. In this case, having the content set and style weight both set to 100, 20000 steps, and a learning rate of 0.003. This resulted in a very similar image to the one where only the step number had changed, but with a loss value of 152.95. This was the lowest loss value reached in this experiment, and the image itself is also a very impressive combination of style and content. No color, proper background, and manga artstyle are all apparent, with the original content completely conserved. Following this, I attempted to see if any difference to content and style weights influenced the final design as much as increasing the step count to 20000 did. Figure 16 shows a high style-weight-to-content-weight ratio (content of 1 and style of $1e9$), with the same learning rate and step number as Figure 15. As can be seen, this had very little effect on the image itself, but the total loss was $1.4e8$. It's fascinating that despite the total losses being completely different, the images appear to be very similar. Finally, I tested the opposite – a high content-weight-to-style-weight ratio (content of $1e4$ and style of 100). Figure 17 is my personal favorite of the bunch, as while the loss is 1676 compared to Figure 15's value of 152.95, I find the touch of color that remains to be a good combination of style and content coloring.

Analysis

From the experiments, several key trends were observed. Firstly, style weight has the strongest influence on visual appearance due to the multiple layers and weights attributed to it. Content weight primarily affects structural preservation, but the ratio between the two is what really matters. More important than that, however, is that the learning rate impacts optimization stability and convergence, but is also closely related to how style is used in the output. Finally,

the most concrete trend is that increasing the number of steps significantly improves output quality, which makes complete sense; more time to converge on the global minima will of course produce the best results. Additionally, total loss values do not always directly correspond to perceptual quality, or even image similarity – highlighting the importance of qualitative evaluation.